

**SIMATS ENGINEERING  
ASSESSMENT TOOL 3**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1708

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**COURSE OUTCOME COVERED**

**CO3:** *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 20%*

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**QUESTIONS: 5**

**Time Duration: 1 Hour  
Total Marks:30**

Annexure A

**Resolution Algorithm Implementation**

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***Code of Conduct:***

*I \_M Kavya 192472370\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: kavya***

## Resolution Algorithm Implementation

### Question 1 – Rain and Wet Ground:

#### ➤ Problem Understanding

The knowledge base contains two statements:

1. If it rains, then the ground becomes wet.
2. It is raining.

The goal is to prove that the ground is wet using the Resolution Algorithm.

#### ➤ Propositional Logic Representation

Let:

- **R** = It is raining.
- **W** = The ground is wet.

The given statements can be represented as:

1. If it rains, then the ground becomes wet:  **$R \rightarrow W$**
2. It is raining: **R**

Goal: **W**

#### ➤ Conversion into CNF

The implication is converted using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$R \rightarrow W \equiv \neg R \vee W$$

The CNF clauses are:

- **C1:  $\neg R \vee W$**
- **C2: R**

#### ➤ Negation of the Goal

Goal: **W**

Negated goal:  **$\neg W$**

Therefore, add the following clause:

- **C3:  $\neg W$**

#### ➤ Resolution Steps

## Step 1

Resolve C1 and C2:

- C1:  $\neg R \vee W$
- C2:  $R$

The complementary literals are  $\neg R$  and  $R$ .

After resolution:  $W$

Therefore:

- C4:  $W$

## Step 2

Resolve C4 and C3:

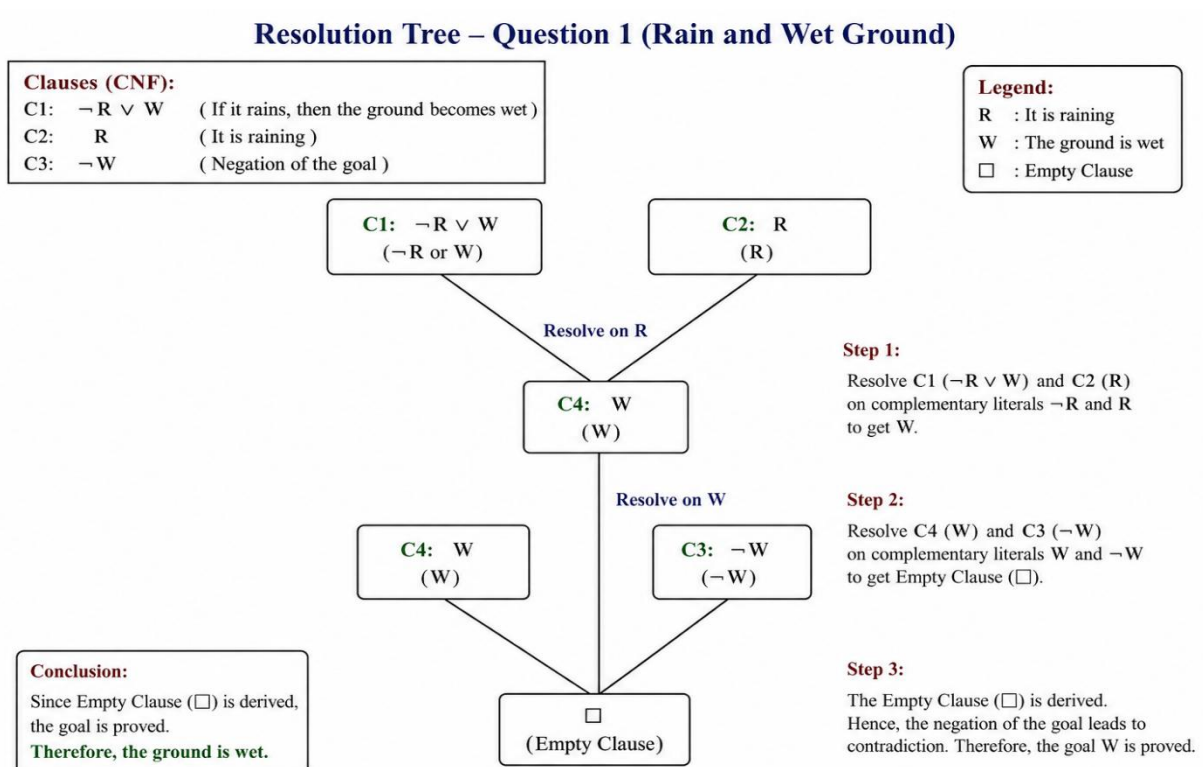
- C4:  $W$
- C3:  $\neg W$

The complementary literals are  $W$  and  $\neg W$ .

After resolution:

The empty clause is obtained.

## 6. Resolution Tree



## 7. Conclusion

Since the resolution process derives the **Empty Clause** ( $\square$ ), the negation of the goal leads to a contradiction.

Therefore, the goal is proved:

**$\therefore$  The ground is wet.**

## **Question 2 – Student Assignment Submission:**

### **1. Problem Understanding**

The knowledge base contains:

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

The goal is to prove that Rahul receives internal marks using the Resolution Algorithm.

### **2. Propositional Logic Representation**

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

The statements are represented as:

1. If Rahul submits the assignment, then Rahul receives internal marks:  **$S \rightarrow M$**
2. Rahul submitted the assignment: **S**

Goal: **M**

### **3. Conversion into CNF**

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Statement 1 becomes:  **$\neg S \vee M$**

Statement 2 is already in CNF: **S**

Therefore, the CNF clauses are:

- **C1:  $\neg S \vee M$**
- **C2: S**

### **4. Negation of the Goal**

Goal: **M**

Negated goal:  **$\neg M$**

Add:

- **C3:  $\neg M$**

### **5. Resolution Steps**

#### **Step 1**

Resolve C1 and C2:

- **C1:  $\neg S \vee M$**
- **C2: S**

The complementary literals are  $\neg S$  and  $S$ .

Therefore: **M**

New clause:

- **C4: M**

## Step 2

Resolve C4 and C3:

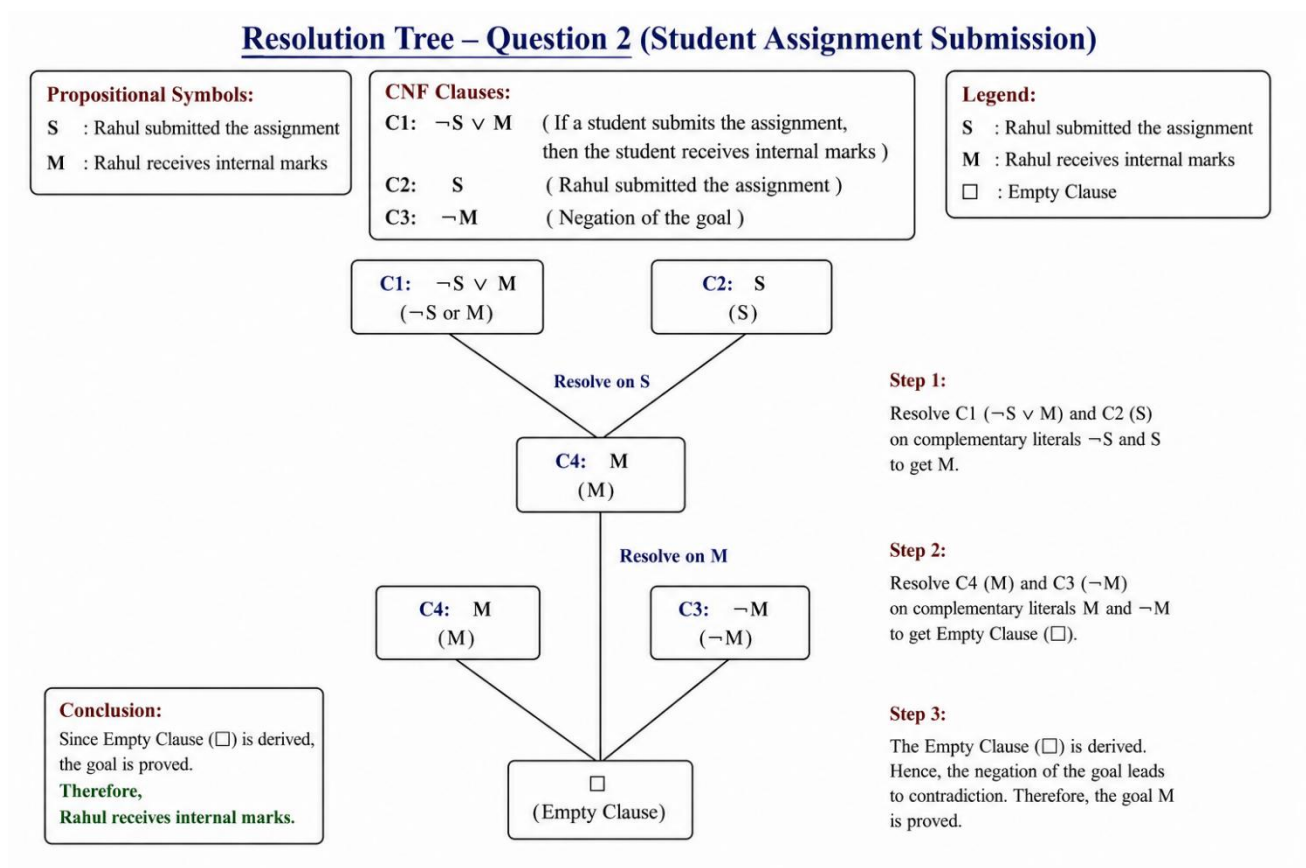
- C4: **M**
- C3:  $\neg M$

The complementary literals are **M** and  $\neg M$ .

Therefore:

The empty clause is obtained.

## 6. Resolution Tree



## 7. Conclusion

Since the Empty Clause ( $\square$ ) is derived, the negation of the goal is contradictory.

Therefore, the goal is proved:

**$\therefore$  Rahul receives internal marks.**

### Question 3 – Library Membership

#### 1. Problem Understanding

The knowledge base contains:

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

The goal is to prove that Priya can borrow books using the Resolution Algorithm.

#### 2. Propositional Logic Representation

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

The statements are:

1. If Priya is a library member, then Priya can borrow books.  **$L \rightarrow B$**
2. Priya is a library member. **L**

Goal: **B**

#### 3. Conversion into CNF

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Statement 1:

$$L \rightarrow B$$

becomes:

$$\neg L \vee B$$

Statement 2:

$$L$$

is already in CNF.

Therefore:

- **C1:  $\neg L \vee B$**
- **C2: L**

#### 4. Negation of the Goal

Goal: **B**

Negated goal:  **$\neg B$**

Add:

- **C3:  $\neg B$**

#### 5. Resolution Steps

##### Step 1

Resolve C1 and C2:

- $C1: \neg L \vee B$
- $C2: L$

The complementary literals are  $\neg L$  and  $L$ .

Therefore:

**B**

New clause:

- **$C4: B$**

## Step 2

Resolve  $C4$  and  $C3$ :

- $C4: B$
- $C3: \neg B$

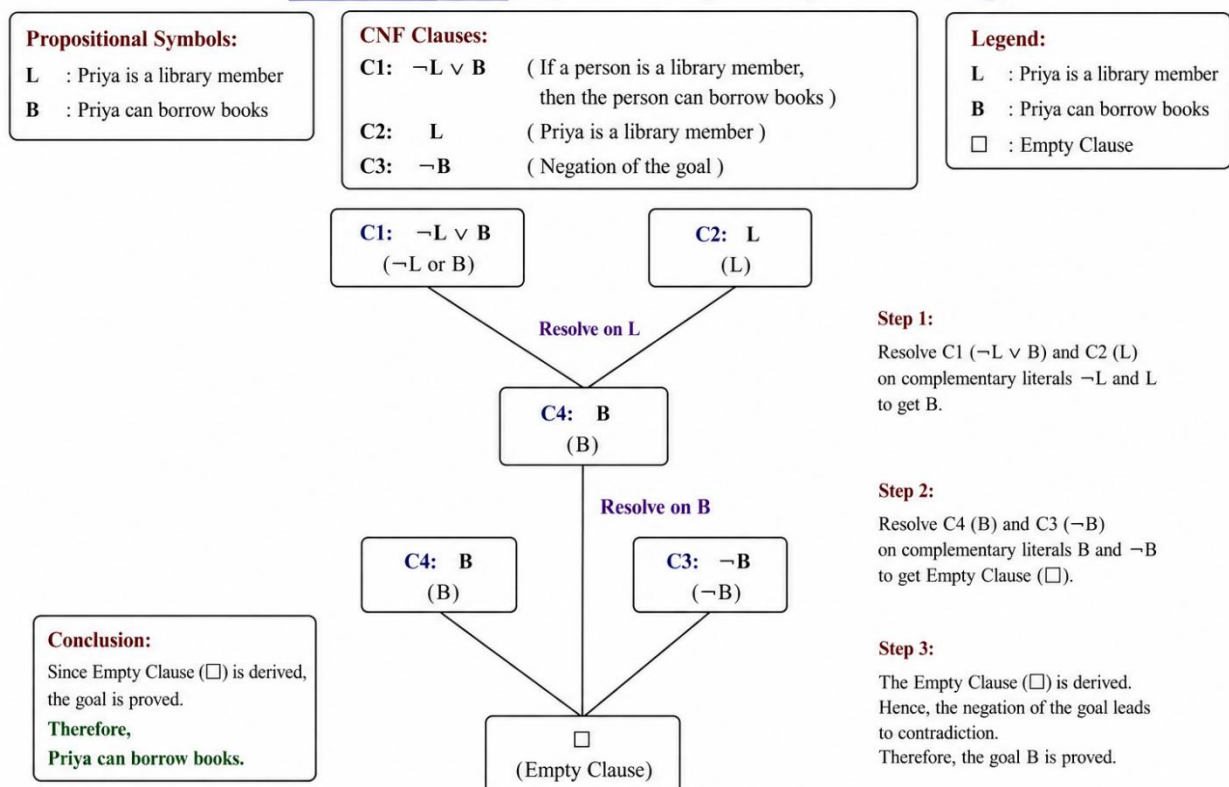
The complementary literals are **B** and  $\neg B$ .

Therefore:

The Empty Clause is obtained.

## 6. Resolution Tree

### Resolution Tree – Question 3 (Library Membership)



## 7. Final Conclusion

The resolution process derives the Empty Clause ( $\square$ ). Therefore, the negation of the goal is inconsistent with the knowledge base.

Hence, the goal is proved:

**$\therefore$  Priya can borrow books.**

## **Question 4 – Placement Eligibility**

### **1. Problem Understanding**

The knowledge base contains:

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

The goal is to prove that Arun is eligible for placement using the Resolution Algorithm.

### **2. Propositional Logic Representation**

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

The statements are:

1. If Arun clears the aptitude test, then Arun is eligible for placement.  **$A \rightarrow E$**
2. Arun cleared the aptitude test. **A**

Goal: **E**

### **3. CNF Conversion**

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Statement 1 becomes:

$$\neg A \vee E$$

Statement 2 is already in CNF: **A**

Therefore, the CNF clauses are:

- **C1:  $\neg A \vee E$**
- **C2: A**

### **4. Negation of the Goal**

Goal: **E**

Negated goal:

$$\neg E$$

Add:

- **C3:  $\neg E$**

### **5. Resolution Algorithm**

#### **Step 1**

Resolve C1 and C2:



- C1:  $\neg A \vee E$
- C2: A

The complementary literals are  $\neg A$  and A.

After resolution: E

New clause:

- C4: E

## Step 2

Resolve C4 and C3:

- C4: E
- C3:  $\neg E$

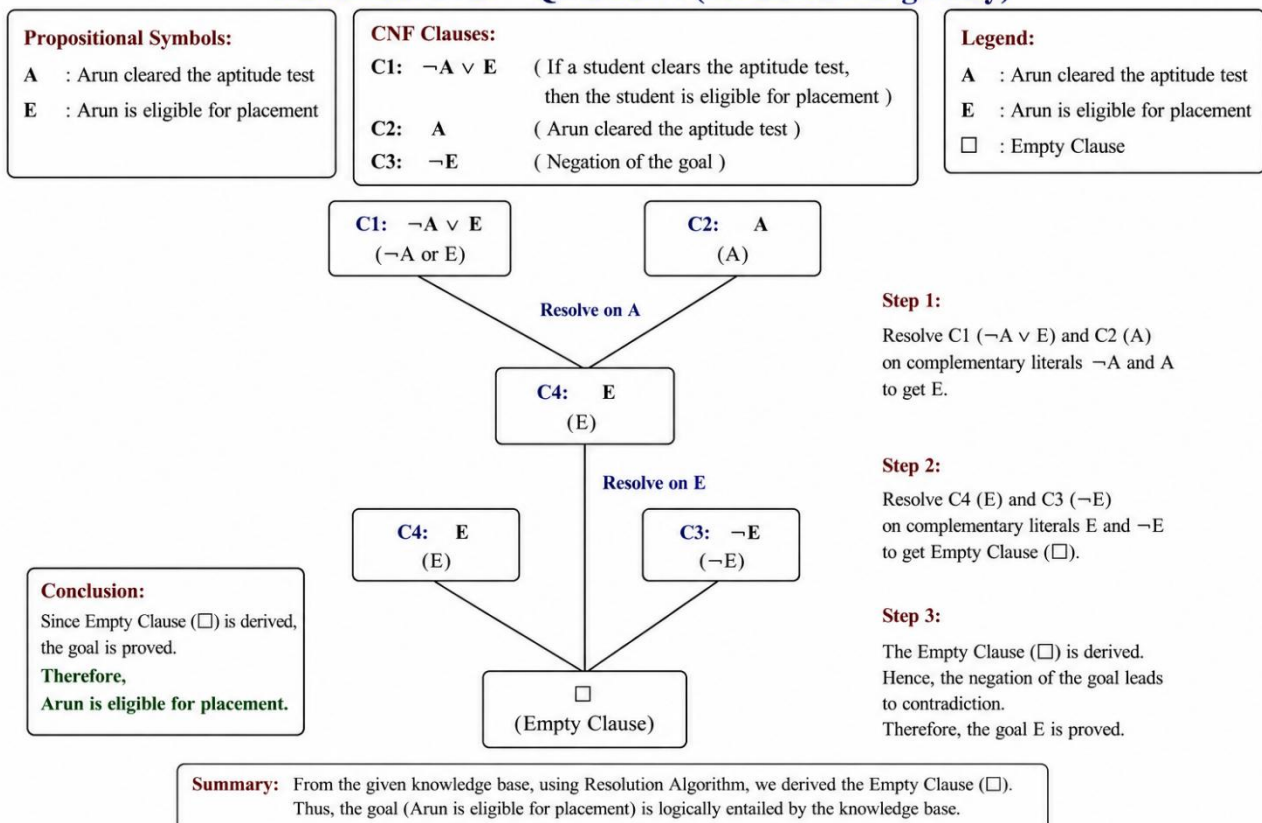
The complementary literals are E and  $\neg E$ .

After resolution:

Thus, the Empty Clause is derived.

## 6. Resolution Tree

### Resolution Tree – Question 4 (Placement Eligibility)



## 7. Final Conclusion

The Empty Clause ( $\square$ ) is obtained through resolution. This means that the negated goal  $\neg E$  contradicts the knowledge base.

Therefore:

**$\therefore$  Arun is eligible for placement.**

## Question 5 – Access Control System

### 1. Problem Understanding

The knowledge base contains three statements:

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

The goal is to prove that the user is granted access using the Resolution Algorithm.

### 2. Propositional Logic Representation

Let:

- **P** = The user entered the correct password.
- **A** = The user is authenticated.
- **G** = The user is granted access.

The statements are:

1. If the user enters the correct password, then the user is authenticated.  $P \rightarrow A$
2. If the user is authenticated, then the user is granted access.  $A \rightarrow G$
3. The user entered the correct password. **P**

Goal: **G**

### 3. Conversion into CNF

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Statement 1:  $P \rightarrow A$

becomes:  $\neg P \vee A$

Statement 2:  $A \rightarrow G$

becomes:  $\neg A \vee G$

Statement 3:

**P**

is already in CNF.

Therefore, the CNF clauses are:

- **C1:**  $\neg P \vee A$
- **C2:**  $\neg A \vee G$
- **C3:** **P**

### 4. Negation of the Goal

Goal: **G**

Negated goal:

**$\neg G$**

Add:

- **C4:  $\neg G$**

## **5. Resolution Steps**

### **Step 1 – Derive Authentication**

Resolve C1 and C3:

- C1:  $\neg P \vee A$
- C3: **P**

The complementary literals are  $\neg P$  and **P**.

Therefore:

**A**

New clause:

- **C5: A**

### **Step 2 – Derive Access**

Resolve C2 and C5:

- C2:  $\neg A \vee G$
- C5: **A**

The complementary literals are  $\neg A$  and **A**.

Therefore:

**G**

New clause:

- **C6: G**

### **Step 3 – Derive Empty Clause**

Resolve C6 and C4:

- C6: **G**
- C4:  $\neg G$

The complementary literals are **G** and  $\neg G$ .

Therefore:

The Empty Clause is obtained.

## 6. Resolution Tree

### Resolution Tree – Question 5 (Access Control System)

#### Propositional Symbols:

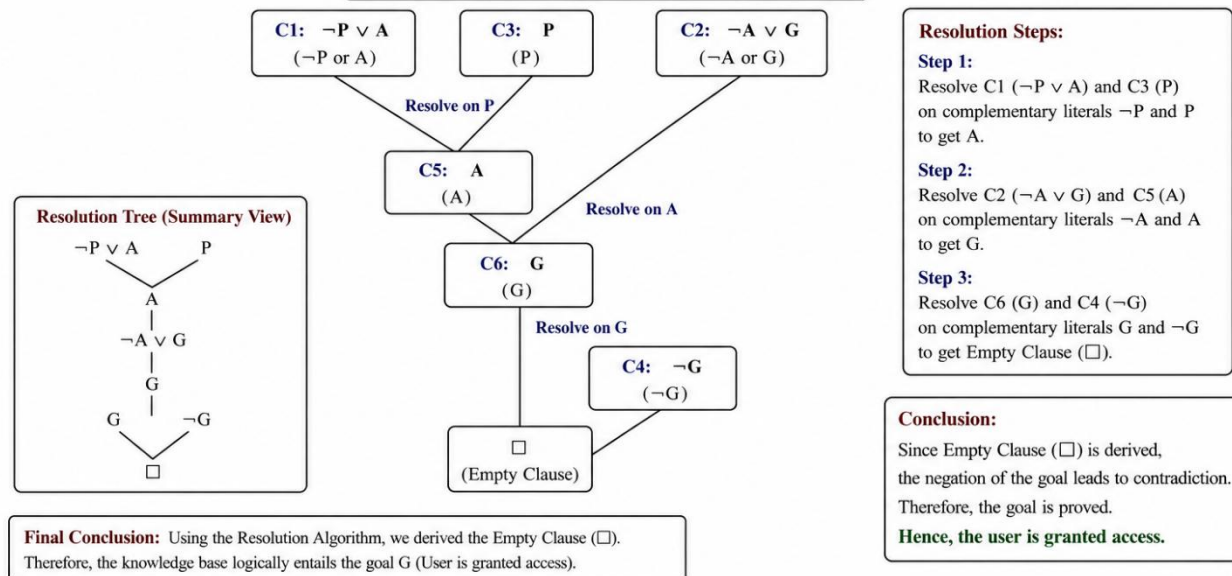
**P** : User enters the correct password  
**A** : User is authenticated  
**G** : User is granted access

#### CNF Clauses:

**C1:**  $\neg P \vee A$  ( If a user enters the correct password, then the user is authenticated )  
**C2:**  $\neg A \vee G$  ( If a user is authenticated, then the user is granted access )  
**C3:** **P** ( The user entered the correct password )  
**C4:**  $\neg G$  ( Negation of the goal )

#### Legend:

**P** : User enters the correct password  
**A** : User is authenticated  
**G** : User is granted access  
 $\square$  : Empty Clause



## 7. Explanation of Resolution Steps

1. From  $\neg P \vee A$  and **P**, the complementary literals  $\neg P$  and **P** are eliminated, giving **A**.
2. From  $\neg A \vee G$  and **A**, the complementary literals  $\neg A$  and **A** are eliminated, giving **G**.
3. The derived clause **G** is resolved with the negated goal  $\neg G$ .
4. This produces the Empty Clause  $\square$ .

## 8. Final Conclusion

Since the Empty Clause ( $\square$ ) is derived, the negation of the goal is contradictory with the knowledge base.

Therefore, the goal is proved:

**∴ The user is granted access.**

## RUBRICS FOR CALCULATION

| Numerical Problems        |           |                                                                                   |                                           |                                            |                                               |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Criteria                  | Weightage | Level 4 (Exemplary)                                                               | Level 3 (Proficient)                      | Level 2 (Developing)                       | Level 1 (Beginning)                           |
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |